

INCIDENT RESPONSE AND MANAGEMENT

What is Incident Response? Incident response refers to deliberate effort cybersecurity analysts take to respond to a cyber attack or cyber threat. Incident response curbs the menace of the attack and further prevents it from happening again.

What is incident management? Incident management refers to stringent actions, policies and protocols a cybersecurity team put in place in order to ensure the prediction, prevention and recovery of important data following a cyber breach.

Key features and differences of Incident Response(IR) and Incident Management(IM)

Incident Response:

A reactive process that focuses on immediate eradication and containment of cyber threats and cyber attack.

IR centers on the identification, quick response and assessing incidents. Some actions involved in IR includes;

- Malware removal
- Isolating compromised systems and devices
- Restoring systems to normal operation

Incident management(IM):

IM is a proactive and reactive process, and it is focused on a broader management activities that includes: the incident lifecycle(preparation, detection, response, containment and eradication, recovery and restoration ,post incident activities)

Differences:

Scope: IR is an immediate response while IM comprises of the entire incident life cycle.

Objective: prioritizes the response to threat and eradicate it, whereas IM aims to minimize impact, restore normal operations and improve the incident handling.

Time lapse: IR is for a short term and occurs when there is an imminent threat, while IM is long-term and progressive.

THE INCIDENT RESPONSE LIFE CYCLE

PREPARATION: the preparation stage is the first area of the incident response life cycle.

Phases of incident response plan(IRP):

- Develop incident Response plan(IRP)
- Establish incident Response Team –
- Conduct risk assessments
- Implement incident detection and response tools
- Develop incident response procedures
- Train and exercise

Step 2: **Establish incident Response Team (IRT).**

- Identify the response team members (e.g security analysts, network administrators analysts).
- Ensure adequate training and expertise and establish clear roles and responsibilities.

Step 3:**Conduct Risk Assessments:**

- Identify the vulnerabilities present and what threat is imminent.
- Assess current security controls and incident response capabilities.

Step 4: **Implement incident Response and detection tools:**

- Installing intrusion detection systems and intrusion prevention systems. These tools work in detecting and preventing breaches in forms of malwares or a change in the network signature.
- Security information and event management (SIEM) systems collect, analyze and monitor log data from various sources to detect and respond to security threats, providing real time insights, alerts and compliance reporting for improved incident response .

Step 6 - Develop incident response procedures:

Incident reporting : Documenting incident details

Initial assessment: Evaluating incident impacts

Notification: Alerting stakeholders and team members

Containment: isolating affected networks

Train and exercise:

- Regular training sessions for IRT members
- Conduct tabletop exercises and simulations
- Test incident response procedures.

DETECTION

Phase 2- **DETECTION**

- This is the second stage of the incident response life cycle, this involves the identification of potential security incidents in real time.

- Steps includes:

- monitoring logs and system activity.
- Analyze network traffic and system behaviour
- Receive alerts from security tools (IDS, IPS, SIEM)
- Receive reports from users and stakeholders.

Detection methods includes:

- Anomaly based detection: this identifies security threats by monitoring deviations from normal system use.

- Signature based detection: this identifies known security threats by matching patterns,hashes,or indicators of compromise (IOC) with predefined threat signatures.
- Behavioural analysis: monitors system and application behaviour to detect and block malicious activities,regardless of known signatures or patterns.
- Machine learning based detection: this uses algorithms to analyse patterns ,identify anomalies and predict potential threats,continuously learning from data and improving detection accuracy over time.
- Human analysis:this involves expert security analysts reviewing and interpreting data, identifying threats, and making informed incident response decisions

DETECTION TOOLS-

- Detection tools:
 - intrusion detection systems
 - security information and event management (SIEM)systems
 - Log management tools
 - Network monitoring tools
 - Endpoint detection and response tools

PHASE 3 - CONTAINMENT

The containment phase aims to limit the spread and impact of a detected security incident.

GOALS:

- Isolate affected systems or networks.
- Prevent further damage or data loss
- minimize incident escalation
- preserve evidence.

Containment strategies:

- Network segmentation
- system isolation
- account lockout
- traffic filtering and firewall rules

Containment

•Containment techniques:

- Disconnecting affected systems from the network
- disabling compromised user accounts
- stopping malicious processes.
- stopping malicious processes
- blocking suspicious network traffick
- creating snapshots for forensic analysis

Containment tools:

- firewalls
- Intrusion prevention systems(IPS)
- Network access control systems.

- Endpoint security solutions
- incident response software

An effective containment mitigates incident impact, allowing for efficient eradication and recovery.

PHASE 4 -ERADICATION

The eradication phase aims at removing the root cause of the security incident ,eliminate the threat and restore systems to a known good state.

GOALS:

- Remove malwares
- Eliminate compromised accounts
- restore compromised systems or data
- fix vulnerabilities exploited by the threat.

Eradication techniques:

- using antivirus software to remove malware
- conducting thorough system scans.

- analyzing and removing suspicious files
- implementing security patches and updates

Eradication tools:

- Antivirus software
- Endpoint security solution
- vulnerability scanners
- system restoration software.

Phase 5-Recovery

The aim of the recovery phase is to restore systems, data and business operations to a normal state after eradication.

GOALS:

- restore system functionality
- recover lost data
- verify system security

TECHNIQUES FOR RECOVERY

- Backup and restore
- data recovery software
- system reconfiguration and rebuilding
- network reconnection and testing.